Form D

Core Focus on Proportions & Probability

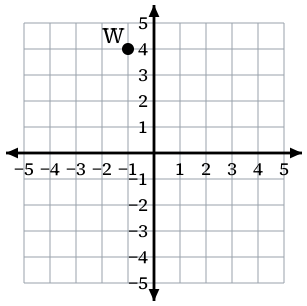
Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

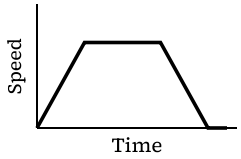
1. Which ordered pair represents the point graphed?



A. (4, −1)
B. (1, −4)
C. (−4, 1)
D. (−1, 4)

D. (−1, 4): start at the origin, move 1 left, then 4 up. (A swaps x and y.)

2. Which story best matches the speed graph below?

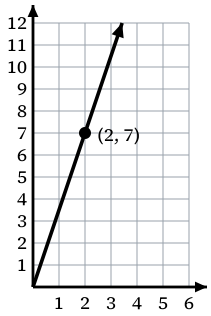


A. Ella walked to the park, sat on a bench, and then walked home.
B. A runner ran at the same speed the whole time.
C. A skateboarder slowed down, stopped, and then sped up again.
D. A bus sped up, drove at a steady speed, then slowed down and stopped at a bus stop.

D. On a speed graph, rising means speeding up, flat means a steady speed, and falling to 0 means slowing to a stop. (A describes a distance graph: the flat part there would mean stopped.)

3. If the graph below models direct variation, what is its rate?

A. $\frac{2}{7}$
B. 3.5
C. 7
D. The graph does not model direct variation.



B. 3.5. The line is straight and passes through (0, 0), so it is direct variation. Rate = y ÷ x = 7 ÷ 2 = 3.5. (2/7 is x ÷ y, the wrong way round.)

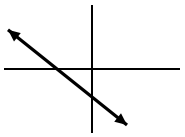
4. Which equation is the direct variation equation for the values in the table below?

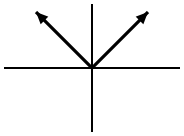
x	4	8	12	16
y	3	6	9	12

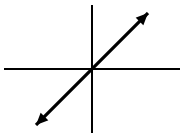
A. y = 3x
B. y = $\frac{4}{3}$x
C. y = $\frac{3}{4}$x
D. y = x − 1

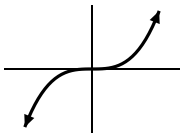
C. y = ¾x (m = y ÷ x = 3 ÷ 4 = ¾; check 6 ÷ 8 = ¾, 9 ÷ 12 = ¾, 12 ÷ 16 = ¾). D fits only the first column: 8 − 1 = 7, not 6.

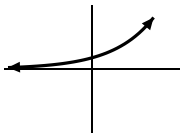
5. Which graphs show direct variation? Circle all that apply.

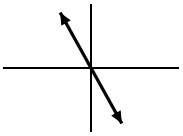
A. 

B. 

C. 

D. 

E. 

F. 

C, F. Each is a straight line through the origin. A is a line that misses the origin; B and D pass through the origin but are not straight lines; E is a curve that misses the origin.

6. Which equations are direct variation equations? Circle all that apply.

A. $y = 8 + x$

B. $y = x$

C. $y = \frac{12}{x}$

D. $y = -2.5x$

E. $y = 4 - 3x$

F. $y = -\frac{x}{2}$

B, D, F. Each has the form y = mx (m = 1, −2.5, −½). A and E add or subtract a number, C has x in the denominator.

7. The table below models direct variation. What is the missing value?

x	0	2	4	...	9
y	0	5	10	...	?

A. 18

B. 19

C. 22.5

D. 45

C. 22.5. Rate = 5 ÷ 2 = 2.5, so y = 2.5x. When x = 9: y = 2.5 × 9 = 22.5.

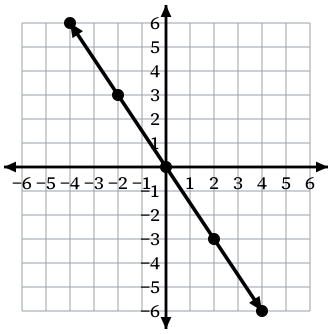
8. What is the direct variation equation for the line graphed below?

A. $y = -\frac{3}{2}x$

B. $y = -\frac{2}{3}x$

C. $y = \frac{3}{2}x$

D. $y = -3x$



A. y = −3/2 x (the line passes through (0, 0) and (2, −3): m = −3 ÷ 2 = −3/2; it falls from left to right, so the slope is negative)

Form D

Core Focus on Proportions & Probability

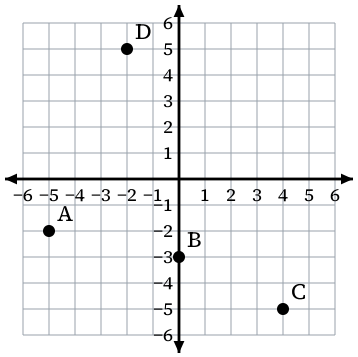
Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part II – Constructed Response

Write the ordered pair for each point on the coordinate plane to the right. Then name the quadrant the point is in, or the axis it is on.



1. A

(−5, −2); Quadrant III

2. B

(0, −3); on the y-axis (in no quadrant)

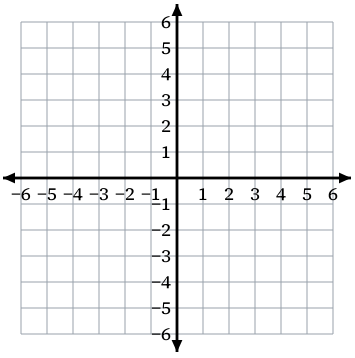
3. C

(4, −5); Quadrant IV

4. D

(−2, 5); Quadrant II

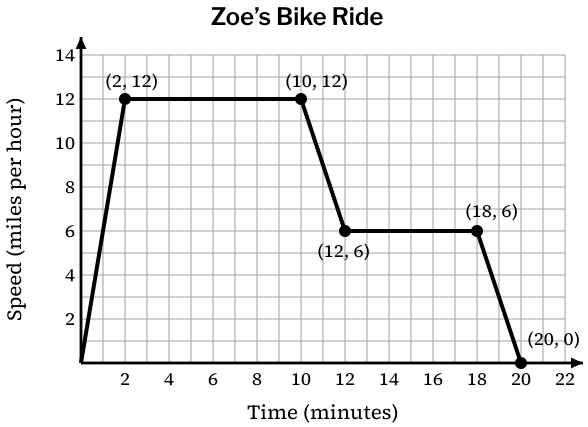
5. The points A(−5, 4), B(1, 4) and C(1, −2) are three corners of square ABCD. Plot A, B and C. Then plot the fourth corner, D, and connect the points in order to make the square. What are the coordinates of D?



D = _____

D(−5, −2). AB goes from x = −5 to x = 1, so it is 6 units long; BC goes from y = 4 to y = −2, also 6 units. D is 6 units straight down from A: (−5, 4 − 6) = (−5, −2).

6. Zoe rode her bike to a friend's house. The graph below shows her **speed** (miles per hour) over time (minutes). For each part, explain how you know your answer is correct.



a. What was Zoe's fastest speed?

12 miles per hour. The highest part of the graph is at 12.

b. For how many minutes did Zoe ride at her fastest speed?

8 minutes. The graph is flat at 12 from minute 2 to minute 10: 10 − 2 = 8.

c. What was happening to Zoe's speed from minute 10 to minute 12?

She slowed down from 12 mph to 6 mph (a drop of 12 − 6 = 6 mph). The graph goes down.

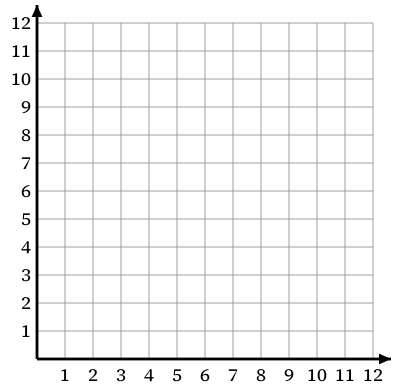
d. What was Zoe doing from minute 12 to minute 18? Was she stopped?

She rode at a steady (constant) speed of 6 mph. She was not stopped: a flat part on a speed graph means constant speed. She stopped at minute 20, where the speed reaches 0.

Complete each input-output table and graph each function in Quadrant I. What is the slope of each direct variation equation?

7.

Input x	Function Rule y = 2.5x	Output y
0		
1		
2		
3		
4		

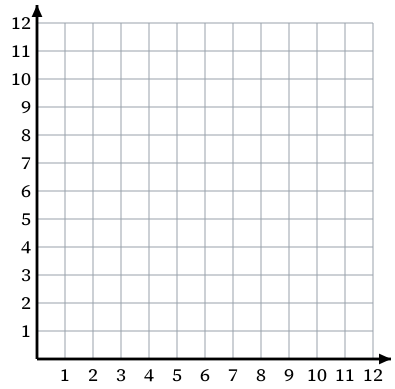


slope: _____

Outputs: 0, 2.5, 5, 7.5, 10 (2.5(0) = 0; 2.5(1) = 2.5; 2.5(2) = 5; 2.5(3) = 7.5; 2.5(4) = 10). Graph: points (0, 0), (1, 2.5), (2, 5), (3, 7.5), (4, 10) on a straight line through the origin. Slope = 2.5.

8.

Input x	Function Rule $y = \frac{3}{4}x$	Output y
0		
4		
8		
12		



slope: _____

Outputs: 0, 3, 6, 9 (¾(0) = 0; ¾(4) = 3; ¾(8) = 6; ¾(12) = 9). Graph: points (0, 0), (4, 3), (8, 6), (12, 9) on a straight line through the origin. Slope = ¾ (or 0.75).

Determine whether each table models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

9.

x	y
−4	−10
−2	−5
2	5
4	10
6	15

direct variation? **YES** **NO**

rate/explanation:

YES; rate 2.5 (or 5/2). y ÷ x = 2.5 for every pair: −10 ÷ (−4), −5 ÷ (−2), 5 ÷ 2, 10 ÷ 4, 15 ÷ 6 all equal 2.5. The table does not list (0, 0), but the line still passes through it.

10.

x	y
−3	9
−1	1
0	0
2	4
4	16

direct variation? **YES** **NO**

rate/explanation:

NO. The ratios y ÷ x are not equal: 9 ÷ (−3) = −3, 1 ÷ (−1) = −1, 4 ÷ 2 = 2, 16 ÷ 4 = 4. The graph passes through (0, 0) but is a curve, not a straight line.

Determine whether each equation models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

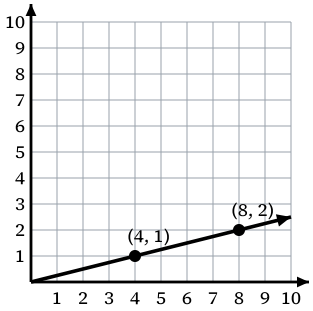
11. $y = -x$ direct variation? **YES** **NO** rate/explanation:

YES; rate −1. y = −x is the same as y = −1x, which has the form y = mx with m = −1.

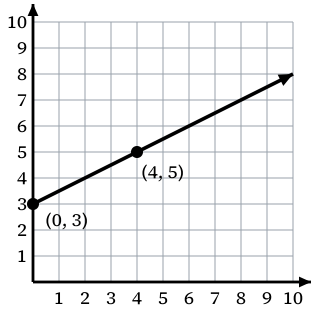
12. $y = \frac{x}{2} + 1$ direct variation? **YES** **NO** rate/explanation:

NO. The "+ 1" means the line crosses the y-axis at (0, 1), not at the origin. (y = x/2 alone would be direct variation.)

13. Omar says Graph A models direct variation and has a rate of 4, because 8 ÷ 2 = 4. Priya says Graph B models direct variation because it is a straight line and has a rate of $\frac{1}{2}$. Is either student correct? Explain each student's mistake.



Graph A



Graph B

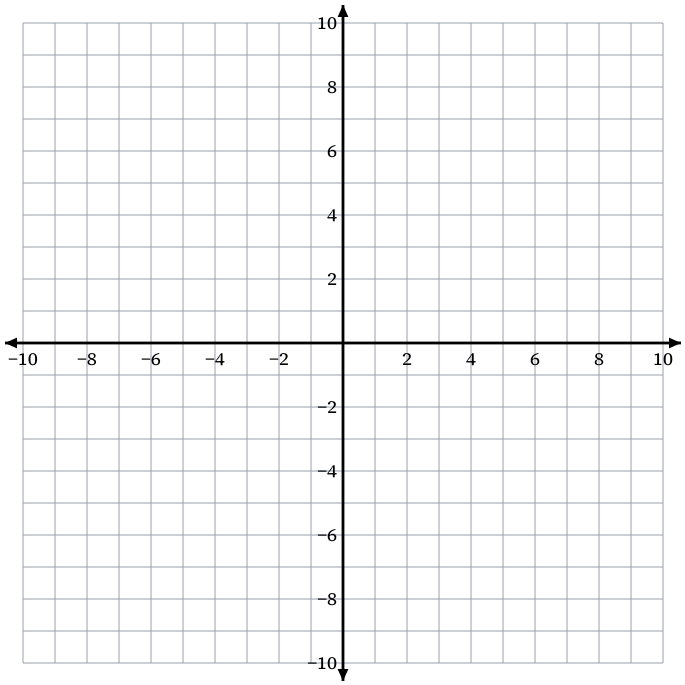
Neither student is fully correct. Omar is right that Graph A is direct variation (a straight line through (0, 0)), but he divided x ÷ y. The rate is y ÷ x = 2 ÷ 8 = 1 ÷ 4 = ¼. Priya is wrong: Graph B is a straight line, but it starts at (0, 3), so it does not pass through the origin and is not direct variation.

14. A machine fills juice bottles at a steady rate. The table below shows the number of bottles filled, y, after x minutes. Write a direct variation equation for the table. Then use it to find how many bottles the machine fills in 25 minutes.

x	y
2	30
5	75
8	120
10	150

y = 15x (m = y ÷ x = 30 ÷ 2 = 75 ÷ 5 = 15). In 25 minutes: y = 15 × 25 = 375 bottles.

15. Graph the direct variation equation $y = -\frac{2}{3}x$.



A straight line through the origin that falls from left to right with slope −⅔ (down 2 for every 3 to the right): it passes through (−9, 6), (−6, 4), (−3, 2), (0, 0), (3, −2), (6, −4), (9, −6), in Quadrants II and IV.

Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part III – Problem Solving

The total cost of bags of mulch varies directly with the number of bags. The table below shows what Theo paid for mulch for his garden.

Bags of Mulch	Total Cost ($)
0	0.00
4	27.00
8	54.00
12	81.00
16	108.00

How many bags of mulch can Theo buy for $148.50?

22 bags. Rate = 27.00 ÷ 4 = $6.75 per bag (54 ÷ 8 = 6.75 too), so y = 6.75x. 148.50 = 6.75x → x = 148.50 ÷ 6.75 = 22. Check: 22 × 6.75 = 148.50.